Winter term 2020/2021: Exercises "Radio Frequency Technology"

EXERCISE SESSION 5 (04.12.2020)

Problem 5.1: Field intensity in dominant mode

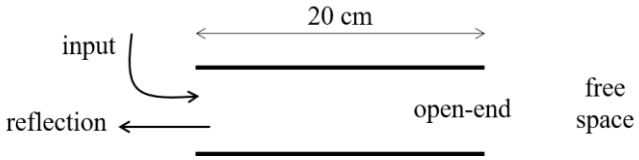
An air-filled loss-less metal rectangular waveguide of dimensions a = 4 cm and b = 2 cm transports a time average power of 2 mW at 10 GHz in its fundamental mode. Calculate the peak value of electric field inside the waveguide.

Problem 5.2: Dispersion of modes in metal waveguides

A air-filled rectangular metal waveguide of width 1 cm and height 3 cm is operating in TE_{12} mode at a frequency that is 20% higher than its corresponding cut-off frequency. Calculate the operating phase velocity and group velocity.

Problem 5.3: Open-end waveguide

A loss-less rectangular metal waveguide of dimensions a waveguide is of a = 4 cm, b = 2 cm and length 20 cm is operating in TM_{21} mode at 20 GHz and is left open to free space in one of its ends (ignore any edge effect). See the figure below. For the input reflection coefficient, calculate the magnitude (dB) and phase (degrees) with respect of the input signal.



Short questions session 5

a) What is the primary boundary condition that must be fulfilled in metallic waveguides?

b) What is the fundamental mode in a rectangular waveguide with a>b? and, if the waveguide is b>a?

c) Calculate the fundamental mode cutoff frequency for a waveguide with a=20 cm and b=10 cm.

d) For the previous waveguide, which is the first higher order mode? Which would it be if the b side is 15 cm?

e) and, what if it is 5 cm?

f) What is the spatial dependency in x, y and z (not *time*-dependency) of each of the electric field components in a TE10 mode?

g) Order the next modes from lower to higher cutoff frequency in a rectangular waveguide with a>b: TE11, TE22, TM10, TM33, TE21, TE12, TE33, TE10

h) A signal is transmitted through the fundamental mode of a rectangular waveguide with a frequency $f_1 = f_0$ above cutoff. The frequency is increased to $f_2 = 2f_0$ and transmitted also in fundamental mode. What would be the new value of λ_{g2}? Bigger, smaller or equal to $\lambda_{g1}/2$? What about λ_{c2}?

i) Which problem could we find in this case?